Solinas number fields

Experiment observations and a worked explanation · 11 October 2026 UTC

EXP-ECDLP-4c6a09 · H-ECDLP-4da36f · GOAL-ECDLP-001

Recorded execution

The existing complete input set finished **1395 cells** in **768.0 s**. A full three-seed replay finished in **823.4 s**. The raw results are **byte-identical**. All recorded cells reached W=100,000. These are observations; the scientific hypothesis has not been promoted.

Shape	Solinas n	Mean γ	Random n	Mean γ	Difference
P-192	32/45	0.057999	217/300	0.074362	−0.016363
P-224	4/15	0.000000	129/300	0.077376	−0.077376
P-256	4/15	0.076352	93/300	0.091570	−0.015218
P-384	1/15	0.000000	40/300	0.045811	−0.045811

“n” gives fitted/total cells. There are **821 missing slopes** overall. Means exclude missing values. Repeated fields and curves are retained; the table does not assert independent sampling.

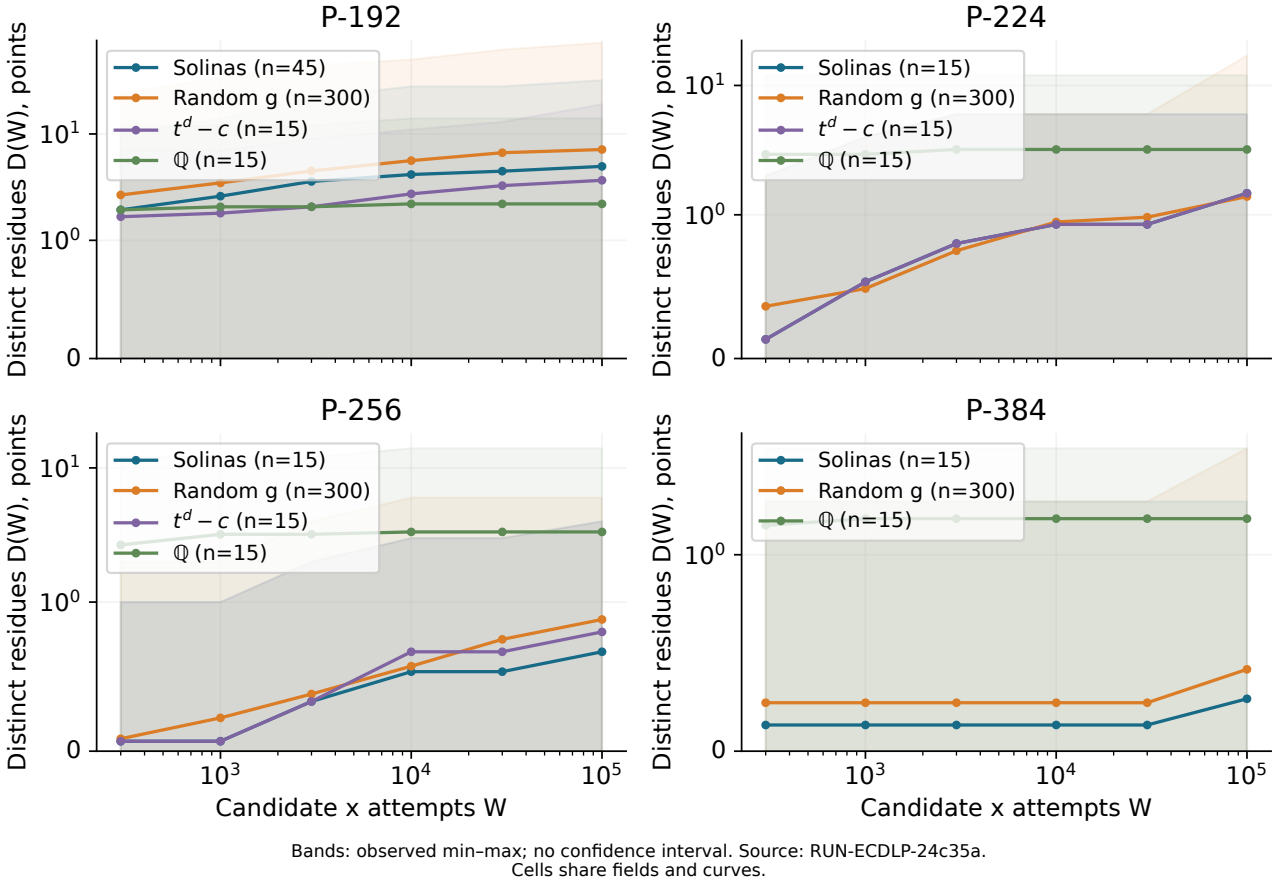
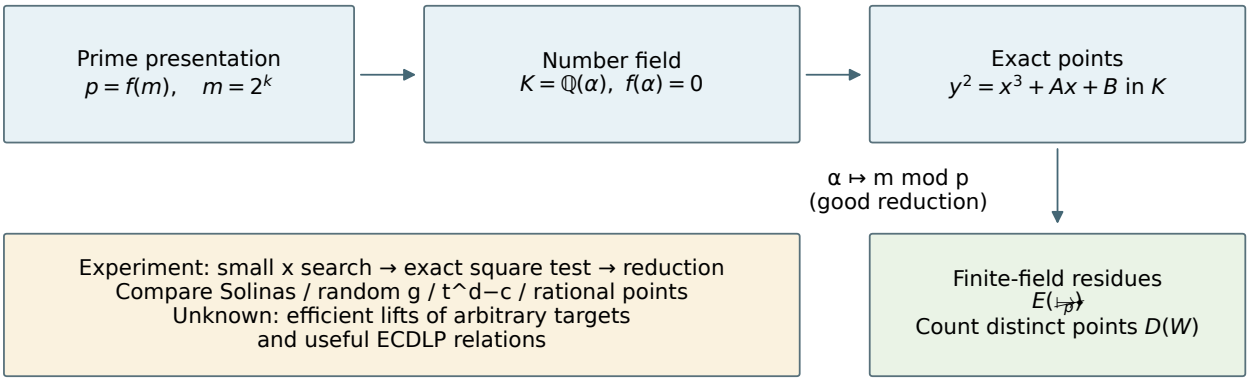


Figure 1: Arithmetic mean coverage across all cells, including zero-point cells. Shading is observed min–max, not a confidence interval.

Primary RUN-ECDLP-24c35a; replay RUN-ECDLP-e09a9d. The two single-threaded invocations overlap on a host with a four-CPU quota.

How the construction works

Derived algebraic construction and measured search; no ECDLP solver is asserted



1. Turn the short prime formula into exact algebra

Write $p = f(m)$, with $m = 2^k$, and introduce a symbolic root α with $f(\alpha) = 0$. If f is irreducible, $K = \mathbb{Q}(\alpha)$ is a number field. For $f(t) = t^3 - t - 1$, this means $\alpha^3 = \alpha + 1$.

2. Reduce the algebra back to the finite field

Because $f(m) = p \equiv 0$ modulo p, substitution $\alpha \mapsto m$ defines a ring map $\mathbb{Z}[\alpha] \to \mathbb{F}_p$. Fractions with invertible denominators can also be reduced. This is not a map from all of the characteristic-zero field K to $\mathbb{F}_p$. At good reduction, elliptic-curve reduction respects point addition.

3. See an actual point identity

Choose $m = 4$, $p = 59$ and $E : y^2 = x^3 - x + 3$. Then $\alpha^3 - \alpha + 3 = 4$, so $(\alpha, 2)$ is an exact point over K. It reduces to $(4, 2)$ because $4^3 - 4 + 3 = 63 \equiv 4$ modulo 59. This is a derived example below the experiment's prime cutoff, not a measured trial or a rank claim. The discriminant $-16 \cdot 239$ is nonzero modulo 59, so this example has good reduction.

4. Ask whether this supplies points efficiently

The experiment tries small power-basis coordinates for x, searches for an exact square root of $x^3 + Ax + B$ in K, then counts distinct reductions. Accepted roots are verified exactly; bounded reconstruction can miss roots. It compares the same search rule in Solinas and random comparison fields.

Why the discriminant and rank matter

The algebraic integers form a lattice with covolume proportional to $\sqrt{|D_K|}$, where D_K is the field discriminant. The root discriminant $|D_K|^{\frac{1}{d}}$ normalizes by degree. Smaller discriminants motivate smaller coordinates; they do not force the curve equation to have a square root. PARI supplies field discriminants, which can differ from the defining-polynomial discriminant.

More discovered points do not certify greater rank: independent generators must be proved independent. The number-field rank arm is explicitly skipped.

What the measurements mean

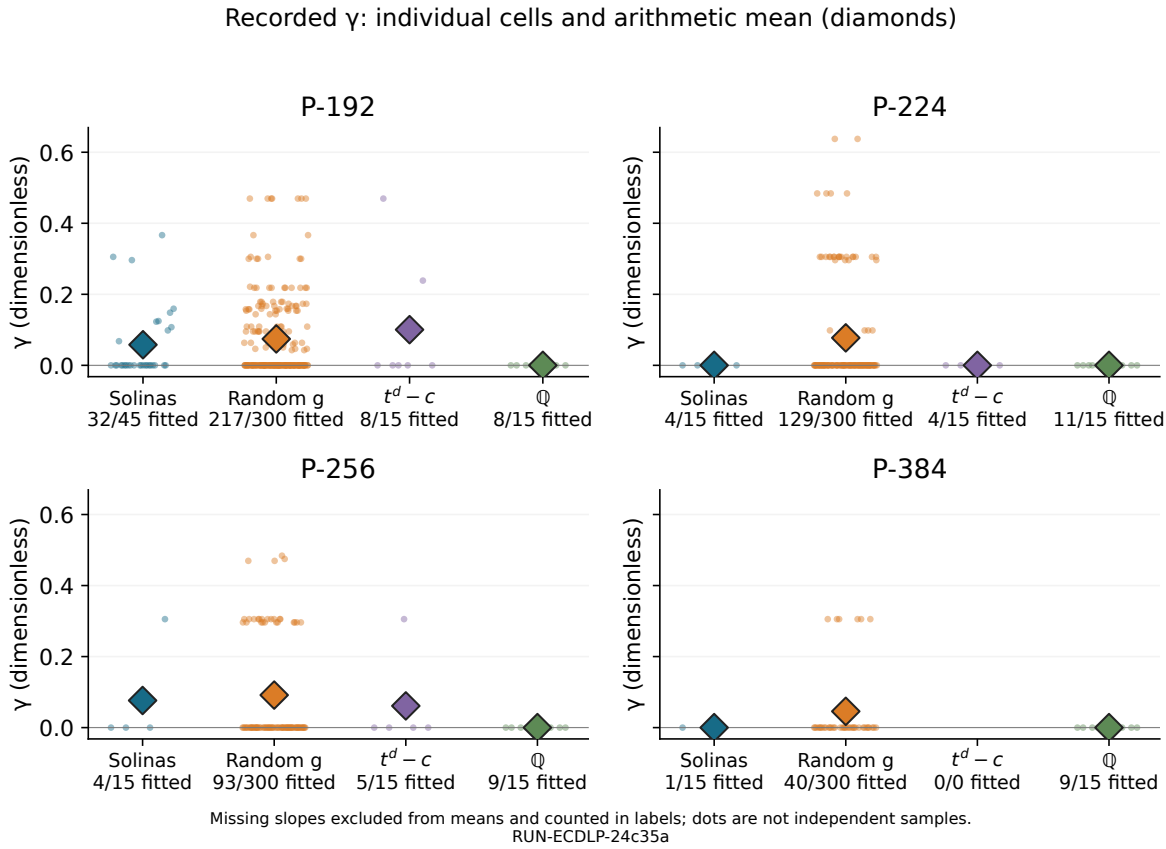


Figure 2: γ is a fitted log–log slope over up to the last three positive-coverage checkpoints. Dots are cell values; diamonds are arithmetic means. This is finite-search growth, not an asymptotic ECDLP exponent.

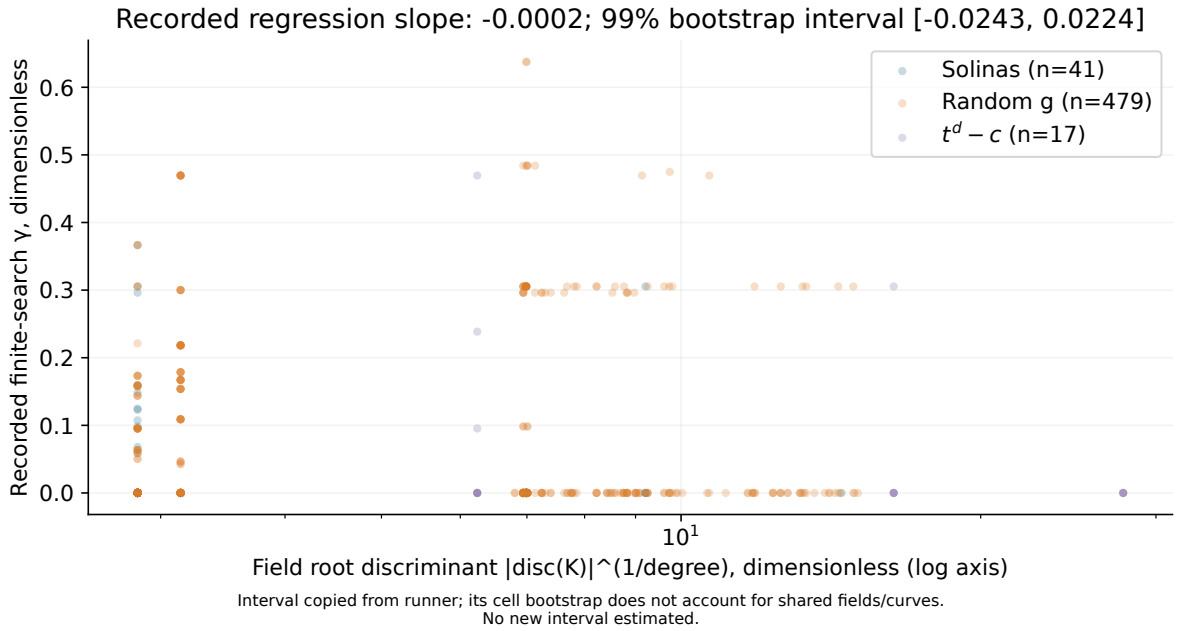


Figure 3: Recorded field discriminants and fitted slopes. The interval is copied from the runner's cell bootstrap; it does not model shared-field or shared-curve dependence.

Recorded regression: slope **−0.000221**, 99% interval **[−0.024314, 0.022378]**, n=537. Non-Q cells without a PARI field discriminant: 0.

Reproduction and remaining gaps

Exact scope

Seeds 1, 2, 3; 20 requested random fields per shape per seed; 5 curves per field; candidate checkpoints 300, 1,000, 3,000, 10,000, 30,000, 100,000; 8 auxiliary primes. Curves are $y^2 = x^3 + Ax + B$ with seeded integer coefficients from −20 to 20 and good reduction at the selected toy prime. The CSV and raw JSON contain every defining polynomial, modulus and curve. No subgroup or unknown discrete-log challenge was selected.

The NIST names label polynomial shapes; this is not a deployed-size curve experiment. Rank and point independence are unmeasured. Heights are logarithmic coefficient-height proxies, not canonical point heights. Random fields match degree and polynomial weight, but can repeat and have different prime sizes.

The gap to a discrete-log algorithm

The multiplicative number field sieve factors small norms into prime ideals to obtain useful log relations. Elliptic-curve points add; small coordinates do not supply the same factorization automatically. A usable method would still need affordable lifts or relations for arbitrary target points, justified independence, and full recovery costs against a baseline. This run measures a supply of points and their coverage, not that method.

Execution identity

Code revision: 8e1cc6c5313fdab887d09b98a712ea05996d7b18

Compiler: rustc 1.99.0 (b940084d7 2026-09-28)

PARI: libpari GP/PARI CALCULATOR Version 2.17.2 (released) (FFI)

The release binary used the complete documented argv. Each process had a 4 GiB address-space cap. Manifests preserve source hashes, command, timestamps and per-cell timing; launch records preserve dirty-tree state. Sources were unchanged. CTRL-SHAPE is True; literal raw-byte CTRL-REPLAY is True. All rank-arm records retain the skipped label.

Evidence and source files

All experiment sources have internal provenance and were inspected in this session. Algebraic identities above are derived. The source implementation is in PR 2035. The complete commands, prime table, counts, required extreme-cell records, resource receipts and caveats are in the accompanying Markdown report.

Primary: RUN-ECDLP-24c35a

Replay: RUN-ECDLP-e09a9d

SHA-256 of raw JSON: 71ffe56253681626967a7b5abf8e9a81dfdc0601a18de3171e21654be8fec835

Primary raw JSON · Manifest · Replay check · Cell identities

The PDF, Typst source and SVG graph sources are communication artifacts. No scientific status, canonical frontier graph or ledger record was changed.